MO'TASEM KASSASBEH

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PROFESSIONAL SUMMARY

High-achieving fourth-year Data Science student (GPA 3.68, Excellent Standing) at the Jordan University of Science and Technology. Possessing a strong foundation in core AI domains, including Machine Learning, NLP, LLMs, and Multi-Agent Systems. Experienced in conducting rigorous academic research—from comprehensive literature synthesis and experimental design to scalable model implementation and benchmarking. Driven by a passion for solving real-world challenges through scientific inquiry, I am eager to transition my academic learning and analytical competencies into impactful, data-driven research contributions.

EDUCATION

Bachelor (B.Sc.) in Data Science — **Jordan University of Science and Technology**

Irbid, Jordan — *October 2022 – June 2026 (Expected)*

Grade: 3.68 / 4.0 **(Excellent Standing)**

Honors: Named to the **Dean's Honor List twice**.

RESEARCH INTERESTS

- **Natural Language Processing (NLP) & Arabic NLP:** Sentiment analysis, fact-checking, and text normalization.
- **Large Language Models (LLMs):** Low-resource fine-tuning, QLoRA, and instruction following architectures.
- **Multi-Agent Systems:** Workflow orchestration, automated content curation, and autonomous agent coordination.
- **Deep Learning Architectures:** CNNs, RNNs/LSTMs, and building neural networks from scratch.

RESEARCH SKILLS

- **Literature Reviewing & Synthesis:** Identifying research gaps and formulating methodologies based on prior work.
- **Experimental Design & Benchmarking:** Setting up rigorous testing environments for model evaluation and comparison.
- **Statistical Data Analysis & EDA:** Thorough exploration, cleaning, and transformation of large-scale datasets.

TECHNICAL SKILLS

Deep Learning: PyTorch, TensorFlow, Neural Nets from Scratch, CNNs (ResNet, DenseNet), RNNs & LSTMs

Generative AI & NLP: LLM Fine-tuning (QLoRA), Transformers (ModernBERT), Multi-Agent (CrewAI), RAG Architectures

Systems & Automation: Workflow Orchestration (n8n), Multi-GPU Training, Linux, Scalable AI Pipelines

Data & Languages: Python, C++, SQL Server, Statistical Segmentation, Benchmarking & EDA

PUBLICATIONS & RESEARCH PAPERS

Arabic Fake News Detection Pipeline Using the AFND Dataset ↗
Ready to submit to ICICS 2026

This project presents the development of an ETL (Extract, Transform, Load) pipeline for processing large-scale Arabic news data. The pipeline integrates articles from multiple sources, merges them with credibility labels, and performs extensive data cleaning including duplicate removal, outlier filtering, and domain-specific Arabic text normalization. Exploratory data analysis (EDA) is conducted to examine dataset distribution, source credibility, and article characteristics. The resulting cleaned and structured dataset of over 360,000 high-quality articles serves as a foundation for future analytics and machine learning tasks, particularly credibility and fake-news detection.

SELECTED PROJECTS

Intelligent ERP System (Graduation Project)
Architecting a next-generation ERP system integrated with AI agents. Automates business workflows, provides intelligent analytics, and features an AutoML pipeline for predictive decision-making.

Career-Hub ↗ Live ↗
An AI-powered platform connecting early-career talent with curated opportunities. Features a Crew AI Multi-Agent System for automated content curation and uses n8n for robust workflow automation.

Arabic Aspect-Based Sentiment Analysis (ABSA) using ModernBERT ↗
Multi-task NLP system for Arabic Food Delivery reviews. Orchestrating aspect detection and sentiment polarity using ModernBERT.

Instruction Fine-Tuning: TinyLlama ↗
Optimizing TinyLlama-1.1B for instruction following using 4-bit quantization (QLoRA) on the databricks-dolly-15k dataset.

Predictive Churn & Segmentation ↗

Dual-stage system integrating K-Means clustering and a DNN (97% accuracy) for automated customer retention strategies.

Sequential Sentiment Architectures ↗

Comparative implementation of Bi-LSTM and Multilayer LSTM architectures for sequential IMDb text data classification.

ResNet-18 From Scratch ↗

Manual implementation of Residual Networks using identity skip connections to solve vanishing gradients on CIFAR-10.

DNN from Scratch (NumPy) ↗

Engineering a fully connected DNN for regression using only NumPy — manual backprop and gradient descent optimization.

CERTIFICATES

- **Transformer-Based NLP Applications** - NVIDIA Deep Learning Institute (DLI) ↗
- **Data Parallelism: Multi-GPU Training** - NVIDIA Deep Learning Institute (DLI) ↗
- **Power BI for Beginners** - Microsoft · Simplilearn ↗
- **Retrieval-Augmented Generation (RAG)** - DeepLearning.AI · Coursera
- **AI Agents** - DeepLearning.AI · Coursera

LANGUAGES

Arabic: Native | **English:** Conversational